

Lecture 3: Population Inference for Means

August 20, 2023

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- ▶ We saw that for large enough sample sizes, our sample proportion was likely to be close to the population proportion.
- ▶ We also demonstrated that we could predict how close we might be by using confidence intervals.
- ▶ We explored a bootstrapping approach to producing confidence intervals.

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- ▶ In this lecture, we consider variables that are not binary and our goal is to learn about the population mean (not just a proportion) from a random sample (with replacement) from the population.
- ▶ Because the variables we will consider may take on many values, we will not know the type of probability distribution underlying our data (e.g., the Bernoulli distribution in the binary variable case).
- ▶ Not knowing the distribution isn't a problem when the sample is large enough, but understanding why it isn't a problem is subtle.

The Current Population Survey

Something about the CPS sampling here...

Recall that our sample mean based on 2,258 working age Californians is \$135,406

Two Questions

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Two Questions

Unlike the turnout example, we will never know the true population income mean, which we will denote with μ . We will also never know the population income standard deviation, which we will denote σ or the population variance which we denote with σ^2 . We also will not be able to generate a sampling distribution. But we still want to answer the two questions.

1. Is \$135,406 likely to be close to the population mean μ ?
2. What can we say about how close?

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

- Recall that we can think of $Y_1, \dots, Y_n$ as *i.i.d.* RVs.
- Although we don't know the distribution $f_Y(y)$ of these RVs, their expected value $E[Y]$ is the population mean μ and their variance $V[Y]$ is the population variance σ^2 .

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

- ▶ Recall that we can think of $Y_1, \dots, Y_n$ as *i.i.d.* RVs.
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What can we learn?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	?
$E(\cdot)$	μ	μ	...	μ	?
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

- ▶ Although we can't observe the population or the sampling distribution, we can mathematically derive certain facts about the sampling distribution.
- ▶ We will focus on the three red question marks.

The Type of Distribution

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	:	:	...	:	:
5	:	:	...	:	:
:	:	:	:	:	:
1 mil	?	?	...	?	?
:	:	:	:	:	:
:	:	:	:	:	:
$f.(\cdot)$	?	?	...	?	$N(?, ?)$
$E.(\cdot)$	μ	μ	...	μ	?
$V.(\cdot)$	σ^2	σ^2	...	σ^2	?

- ▶ The central limit theorem says the sampling distribution for the sample mean will be normally distributed.
- ▶ We know that $\bar{Y}_{2258} \sim N(?, ?)$, but still don't know the mean or the variance of this distribution.

What is the $E[\bar{Y}_n]$?

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(.)$	?	?	...	?	$N(?, ?)$
$E.(.)$	μ	μ	...	μ	?
$V.(.)$	σ^2	σ^2	...	σ^2	?

What is the $E[\bar{Y}_n]$?

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3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(.)$	?	?	...	?	$N(?, ?)$
$E.(.)$	μ	μ	...	μ	?
$V.(.)$	σ^2	σ^2	...	σ^2	?

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean, $E[\bar{Y}] = \mu$.

Proof:

$$\begin{aligned} E[\bar{Y}] &= E\left[\frac{1}{n}(Y_1 + \dots + Y_n)\right] \\ &= \\ &= \\ &= \\ &= \\ &= \mu. \end{aligned}$$

Sample mean is unbiased for the population mean

i	1	2	...	2258	
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5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	$N(\mu, ?)$
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

What is the $V[\bar{Y}_n]$?

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s	Y_1	Y_2	...	Y_{2258}	$\bar{Y}_{2258}$
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5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f(\cdot)$	?	?	...	?	$N(\mu, ?)$
$E(\cdot)$	μ	μ	...	μ	μ
$V(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #2: The *sampling variance* of the sample mean,

$$V[\bar{Y}] = \frac{\sigma^2}{n}$$

Proof:

$$\begin{aligned} V[\bar{Y}] &= V \left[\left(\frac{1}{n} Y_1 + \dots + \frac{1}{n} Y_n \right) \right] \\ &= \\ &= \\ &= \\ &= \\ &= \frac{\sigma^2}{n}. \end{aligned}$$

How much does $\bar{Y}_{2258}$ vary from sample to sample?

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2	?	?	...	?	?
3	?	?	...	?	?
4	:	:	...	:	:
5	:	:	...	:	:
:	:	:	:	:	:
1 mil	?	?	...	?	?
:	:	:	:	:	:
:	:	:	:	:	:
$f.(.)$	?	?	...	?	$N(\mu, \frac{\sigma^2}{n})$
$E.(.)$	μ	μ	...	μ	μ
$V.(.)$	σ^2	σ^2	...	σ^2	$\frac{\sigma^2}{n}$

Facts about the sampling distribution of $\bar{Y}_n$

What we know about the sampling distribution of $\bar{Y}_n$:

1. Approximately normally distributed
2. Centered at μ .
3. Variance of $\frac{\sigma^2}{n}$

These facts imply that the middle 95% of the distribution lies approximately between $\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$ and $\mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$. As long as $\sqrt{n}$ is big in relation to σ , $\bar{Y}_n$ will likely be close to μ .

We still don't know:

1. The value of μ .
2. The value of σ .

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2. Can say something about how close?

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2. Can say something about how close? Let's use a confidence intervals

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